

Quantitative Research Methods
Causal Inference from Observational Data
Advanced module A10 — optional self-study

Prof. Dr. Christoph Weisser

Where this module fits

Course chapter / module	What this module builds on it
Ch. 3 — linear regression	Adjustment <i>is</i> multiple regression; omitted-variable bias
Ch. 4 — logistic regression	The propensity score is a fitted logistic model
Ch. 5 — resampling	Seeded simulations that know the true effect
Module A1 — RCTs	Potential outcomes, selection bias — assumed throughout

The question A1 left open

Module A1 ended with the one clean design: randomise, then compare. This module asks what you can still learn when **nobody randomised** — the data are observational, treatment was *chosen*, and every naive comparison is a mixture of effect and selection.

One of twelve optional advanced modules; self-study. Every number on these slides is reproduced by `make_figures.py` in the module folder.

Contents

- 1 The Problem
- 2 Omitted-Variable Bias
- 3 Adjustment & Matching
- 4 Difference-in-Differences
- 5 Instrumental Variables
- 6 Python Lab
- 7 Summary

Optional and advanced material is collected in the **appendix** at the end of the deck.

Notation in this module

Symbol	Meaning
$D \in \{0, 1\}$	treatment received (not assigned — nobody assigned it)
$Y(1), Y(0)$	potential outcomes with and without treatment (module A1)
U	a confounder: causes both D and Y
X	observed covariates (possibly a noisy proxy for U)
$e(x) = \Pr(D = 1 \mid X = x)$	the propensity score
Z	an instrument: shifts D , touches Y only through D
τ	the causal effect we are after (ATT/ATE)
δ	auxiliary-regression coefficient of D on the confounder

Sources and references

Primary textbook

James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning, with Applications in Python*, Springer. — The regression and classification machinery this module reuses.

Canonical further reading (optional)

Angrist & Pischke, *Mostly Harmless Econometrics* (2009); Hernán & Robins, *Causal Inference: What If* (2020) — both develop everything here at full depth.

Learning objectives

By the end of this module you will be able to:

- **Explain** why a naive group comparison in observational data mixes the causal effect with selection bias.
- **Derive** the omitted-variable-bias formula and predict the sign and size of the bias.
- **Adjust** for observed confounders by regression, stratification and propensity-score matching — and say what adjustment can never fix.
- **Apply** difference-in-differences to a two-group, two-period table and state the parallel-trends assumption.
- **Use** an instrumental variable via the Wald estimator, and check the three conditions an instrument must satisfy.

Roadmap of this module

From “the data say” to “the design says”

- ① The problem: selection, confounding, and one running simulation.
- ② Omitted-variable bias: the formula behind every headline gone wrong.
- ③ Adjustment: regression, stratification, propensity-score matching.
- ④ Difference-in-differences: using time as the control.
- ⑤ Instrumental variables: borrowing someone else's randomness.
- ⑥ Python lab, summary, appendix.

Outline

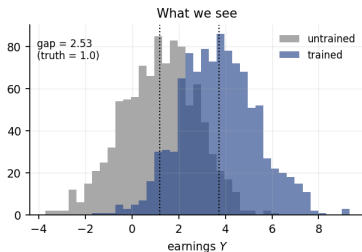
- 1 The Problem
- 2 Omitted-Variable Bias
- 3 Adjustment & Matching
- 4 Difference-in-Differences
- 5 Instrumental Variables
- 6 Python Lab
- 7 Summary

The running example: a job-training programme

A firm offers a voluntary training programme. Ability U drives both **who signs up** (D) and **later earnings** (Y):

$$Y = 2 + \underbrace{1.0}_{\text{true effect}} \cdot D + 1.5 U + \varepsilon, \quad D = 1\{0.9 U + \eta > 0\}, \quad n = 2000.$$

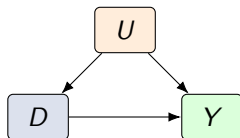
true effect



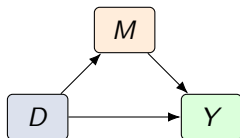
The gap is not the effect

The naive difference in means is 2.53; the true effect is 1.0. The right panel shows why: the trained were more able *before* training.

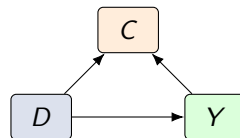
Confounder, mediator, collider — three roles, one picture



confounder: adjust



mediator: do *not* adjust



collider: do *not* adjust

How to read this

Arrows are causes. A **confounder** sits *upstream* of both D and Y and must be adjusted away. A **mediator** carries the effect — adjusting for it deletes the very thing you measure. A **collider** is caused by both — adjusting for it *creates* spurious association. So “control for everything” is not a strategy: what to adjust for is a *causal* decision, made from the diagram, not from p -values.

Exercise A10.1 — Adjust, or leave alone? [Concept]

Exercise

You study the effect of **smoking** (D) on **heart disease** (Y). For each variable, say whether adjusting for it helps, is harmless, or actively biases the estimate — and name its role:

- 1 **Age** — older people smoke more (in this cohort) and have more heart disease.
- 2 **Tar deposits in the lungs** — caused by smoking; contribute to disease.
- 3 **Hospitalisation** — both heavy smokers and heart patients are over-represented in a sample recruited *at a hospital*.

Solution — Exercise A10.1

Solution

Method. Place each variable in the diagram relative to D and Y ; the position decides.

- 1 **Age: confounder** ($\text{Age} \rightarrow D, \text{Age} \rightarrow Y$). Adjust — without it the smoking effect absorbs the age effect.
- 2 **Tar: mediator** ($D \rightarrow \text{Tar} \rightarrow Y$). Do *not* adjust: comparing smokers and non-smokers *at the same tar level* removes the pathway through which smoking does its damage, biasing the total effect toward zero.
- 3 **Hospitalisation: collider** ($D \rightarrow H \leftarrow Y$). Recruiting only hospital patients *conditions* on H and manufactures a (typically negative) spurious association — smokers in hospital are there partly *because* they smoke, patients without heart disease partly because of other illness.

Common mistake

Adding every available covariate to the regression “to be safe”. Item 2 and item 3 are both cases where more adjustment gives a *worse* answer.

Outline

- 1 The Problem
- 2 Omitted-Variable Bias**
- 3 Adjustment & Matching
- 4 Difference-in-Differences
- 5 Instrumental Variables
- 6 Python Lab
- 7 Summary

The omitted-variable-bias formula

Suppose the truth is the **long** regression, but you fit the **short** one:

$$\text{long: } Y = \beta_0 + \tau D + \gamma U + \varepsilon$$

$$\text{short: } Y = b_0 + b_D D + e$$

Rule

$$b_D = \tau + \gamma \delta$$

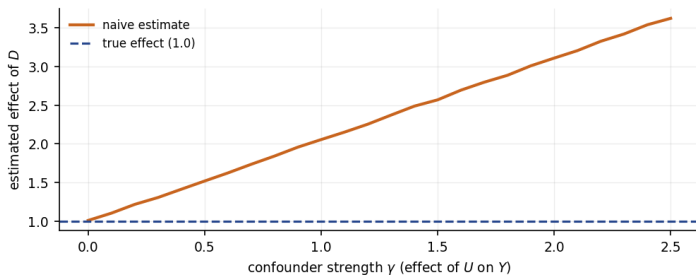
where $U = \delta_0 + \delta D + v$ (the auxiliary regression).

How to read this

The bias is a *product*: how strongly the omitted variable drives the outcome (γ) $\times$ how different the groups are on it (δ). Either factor zero $\Rightarrow$ no bias. Both positive $\Rightarrow$ you *overstate* the effect.

Chapter 3 showed coefficients move when predictors enter; this formula says by exactly how much — for variables you cannot even observe.

The bias is linear in the confounder's strength



Read off the simulation

Dialling the confounder's effect γ from 0 to 2.5, the naive estimate climbs linearly with slope ≈ 1.05 — which is δ , the ability gap between the self-selected groups. At $\gamma = 1.5$: $1.0 + 1.5 \times 1.05 \approx 2.5$, the naive gap we saw.

Exercise A10.2 — Sign and size the bias [Math]

Exercise

A study of **private tutoring** (D) on **exam scores** (Y , points) omits **parental income** U (in \$10k). Assume the long model $Y = \beta_0 + \tau D + \gamma U + \varepsilon$ with $\tau = 5$, $\gamma = 1.5$, and the auxiliary regression $U = \delta_0 + \delta D + v$ with $\delta = 0.8$.

- 1 What does the short regression of Y on D estimate?
- 2 A second omitted variable, **prior struggle**, has $\gamma = -2$ and $\delta = 1.2$ (struggling students get tutors). Its bias?
- 3 With *both* omitted, can you sign the total bias without a computer?

Solution — Exercise A10.2

Solution

Method. Apply $b_D = \tau + \sum_j \gamma_j \delta_j$, one omitted variable at a time; biases from independent confounders add.

- ① $b_D = 5 + 1.5 \times 0.8 = 5 + 1.2 = 6.2$ — tutoring looks 24% better than it is, because richer families buy more of it.
- ② Bias = $\gamma\delta = (-2)(1.2) = -2.4$ — struggling students select *into* tutoring, dragging the tutored group's scores down.
- ③ Total: $5 + 1.2 - 2.4 = 3.8$. The two selections push in *opposite* directions, and here the negative one wins — naive comparisons can just as easily *understate* an effect.

Take-away. You rarely know γ and δ ; the formula's real use is a *sensitivity argument*: “how strong would a confounder have to be to explain my estimate away?”

Outline

- 1 The Problem
- 2 Omitted-Variable Bias
- 3 Adjustment & Matching**
- 4 Difference-in-Differences
- 5 Instrumental Variables
- 6 Python Lab
- 7 Summary

Strategy 1: regression adjustment — and its honest limit

Rule

If **all** confounders are observed as X (*unconfoundedness*), regressing Y on D and X recovers the effect: $\hat{\tau}$ = coefficient of D in $Y \sim D + X$.

The running simulation, adjusted

We observe only a noisy proxy $X = U + \text{noise}$. Adjusting for it moves the estimate from the naive 2.53 to 1.35 — much closer to the truth 1.0, but *not there*: the noise in the proxy leaves residual confounding behind.

The assumption does all the work

Adjustment removes exactly the bias carried by **what you measured**. The claim “no unmeasured confounders” is not testable from the data — it is an argument you must make about the world.

Strategy 2: stratify — compare like with like

The idea in one sentence

Cut the data into groups (strata) inside which X is essentially constant, compare treated and control *within* each stratum, then average the within-stratum gaps.

Why it works

Within a stratum, treated and control have the same X — so the $\gamma\delta$ term of the bias formula is switched off (for the measured X). The naive comparison is a *single* stratum: everyone at once.

Simpson's warning

The within-stratum gaps and the pooled gap can even have *opposite signs* — module A1's Simpson slide is exactly this phenomenon. Stratification is the cure, when the strata are confounders.

Exercise A10.3 — Stratification by hand [Math]

Exercise

A voluntary coaching programme; $X \in \{\text{low}, \text{high}\}$ prior skill. Mean outcomes (group sizes in parentheses):

	coached	not coached
low skill	5.0 ($n = 20$)	4.0 ($n = 80$)
high skill	9.0 ($n = 80$)	8.0 ($n = 20$)

- 1 Compute the naive difference in means (pool the strata).
- 2 Compute the stratified estimate (average the within-stratum gaps, weighting each stratum by its total size).
- 3 Explain the gap between the two numbers in one sentence.

Solution — Exercise A10.3

Solution

Method. Naive = overall treated mean – overall control mean; stratified = size-weighted average of within-stratum gaps.

Part 1 — naive. Treated: $(20 \times 5 + 80 \times 9)/100 = 8.2$. Control: $(80 \times 4 + 20 \times 8)/100 = 4.8$. Naive gap = $8.2 - 4.8 = 3.4$.

Part 2 — stratified. Within low: $5 - 4 = 1$. Within high: $9 - 8 = 1$. Both strata hold 100 people, so the weighted average is 1.0.

Part 3. The coached are mostly high-skill (80 of 100) and the uncoached mostly low-skill, so the naive gap adds the *skill* gap (≈ 4 points, diluted to 2.4) on top of the *coaching* gap (1).

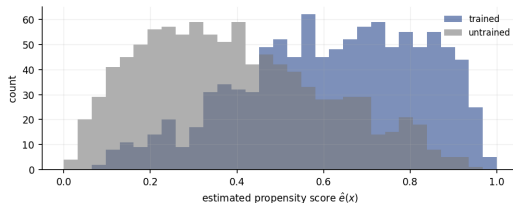
Common mistake

Weighting the within-stratum gaps by the number of *treated* only when you want the ATE — that weighting answers the ATT question instead (here both give 1.0 because the gaps are equal; in general they differ).

Strategy 3: the propensity score — one number replaces all of X

Rule (Rosenbaum & Rubin)

Let $e(x) = \Pr(D = 1 \mid X = x)$, fitted with Chapter 4's logistic regression. If unconfoundedness holds given X , it also holds given the *scalar* $e(X)$ — so it is enough to compare units with the same propensity score.

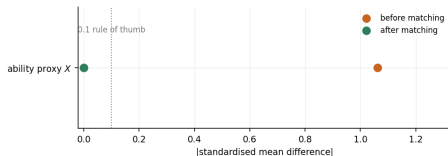


Check overlap before anything else

Matching needs treated and control units at *comparable* scores. Where one histogram is empty, no method can compare — those units have no counterpart.

Propensity-score matching on the running example

Recipe (ATT): for each treated unit, find the control with the nearest $\hat{e}(x)$ (with replacement); average the within-pair differences.



What matching bought us

Before matching the groups differ by 1.06 standard deviations on the ability proxy; after matching, 0.00. The estimate moves from the naive 2.53 to 1.42 — the same story as regression adjustment (1.35), because both lean on the same assumption.

Common mistake

Reporting the matched estimate without the balance picture — the SMD-before/after plot is the evidence that matching did anything.

Adjustment scoreboard — one simulation, four answers

Estimator	Estimate	Why it lands there
Truth (built into the simulation)	1.00	—
Naive difference in means	2.53	effect + selection ($\gamma\delta$)
Regression on D and proxy X	1.35	removes measured confounding only
Propensity matching on X	1.42	same assumption, same residual gap

Adjustment methods agree with each other, not with the truth

Regression and matching answer the *same* question two ways — so their agreement is a consistency check, **not** a validation. Both still carry the confounding that the noisy proxy could not see.

Outline

- 1 The Problem
- 2 Omitted-Variable Bias
- 3 Adjustment & Matching
- 4 Difference-in-Differences**
- 5 Instrumental Variables
- 6 Python Lab
- 7 Summary

Using time as the control group

A policy hits one group (treated) and not another (control); you observe both **before** and **after**.

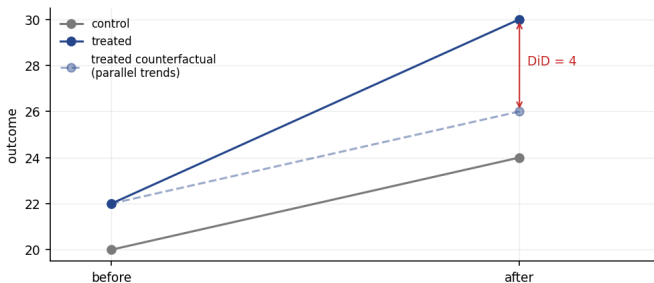
Rule

$$\hat{\tau}_{\text{DiD}} = (\bar{Y}_{\text{T,after}} - \bar{Y}_{\text{T,before}}) - (\bar{Y}_{\text{C,after}} - \bar{Y}_{\text{C,before}})$$

How to read this

The control group's change estimates what would have happened to the treated *anyway* (time trend, seasonality, the economy). Subtracting it removes any confounder that is **constant over time within group** — even unmeasured ones. What it cannot remove: anything that changed for one group only.

DiD in one picture



Parallel trends is the whole bet

Control rose $20 \rightarrow 24$; treated rose $22 \rightarrow 30$; $\text{DiD} = 8 - 4 = 4$. The dashed line is the assumption: untreated, the treated would have risen by the control's $+4$. Plot pre-period trends — parallel *before* is the only evidence you will ever get.

Exercise A10.4 — DiD by hand [Math]

Exercise

A minimum-wage rise hits state T; neighbouring state C is untouched. Average monthly employment per fast-food store:

	before	after
state T (treated)	48	60
state C (control)	50	56

- 1 Compute the DiD estimate of the policy effect.
- 2 Compute the two *wrong* answers: after-minus-before for T only, and T-minus-C after only. What does each one confound the effect with?
- 3 Name one concrete event that would break parallel trends here.

Solution — Exercise A10.4

Solution

Part 1. $(60 - 48) - (56 - 50) = 12 - 6 = +6$ jobs per store.

Part 2. *Before-after in T only:* +12 — confounds the policy with the common time trend (the economy grew: C rose +6 with no policy at all). *T vs C after only:* $60 - 56 = +4$ — confounds the policy with the *permanent* level difference between the states (T started 2 below C, so this understates).

Part 3. Any asymmetric shock: a factory opening in state T between the waves, a tourism collapse hitting only C's border towns, T announcing the rise early so stores hired ahead. Parallel trends is a claim about *events*, not about data.

Common mistake

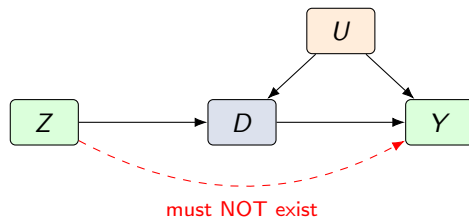
Treating DiD as assumption-free because it “differences everything out”. It differences out *time-constant* confounding only — the two wrong answers in part 2 are each one of the two things it removes.

Outline

- 1 The Problem
- 2 Omitted-Variable Bias
- 3 Adjustment & Matching
- 4 Difference-in-Differences
- 5 Instrumental Variables**
- 6 Python Lab
- 7 Summary

Borrowing someone else's randomness

When a confounder is unmeasured and untimed, one door remains: a variable Z that *nudges* treatment but has no path of its own to the outcome.



The three conditions

- 1 **Relevance:** Z actually moves D (testable — the first stage).
- 2 **Exclusion:** Z affects Y *only* through D (not testable).
- 3 **Independence:** Z is as good as random — unrelated to U (not testable).

The Wald estimator — two small divisions

Rule (binary Z)

$$\hat{\tau}_{IV} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}}$$

effect of the nudge on the outcome
effect of the nudge on take-up

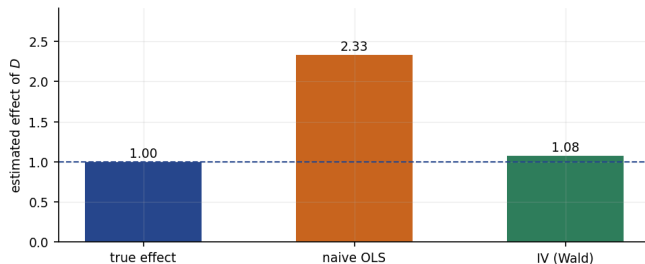
How to read this

The numerator is an *intention-to-treat* effect (module A1); the denominator is the **first stage**. Dividing scales the ITT up to the effect on those the nudge actually moved — the *compliers*. This is the same ITT logic A1 used for non-compliance, now run in reverse.

The running simulation, instrumented

A random encouragement Z raises take-up by 0.40 (first stage). Naive OLS: 2.33. Wald: 1.08 — on top of the truth 1.0, with the confounder U still completely unobserved.

Three estimators, one target



Why IV wins here — and what it costs

IV never observed the confounder, yet lands on the truth: the encouragement was *randomised*, so its variation in D carries no selection. The price: a **weak first stage** blows the variance up ($1/0.40$ scales every error), and the answer is the compliers' effect, not everyone's.

Exercise A10.5 — Is it an instrument? [Concept]

Exercise

You want the effect of **university attendance** (D) on **earnings** (Y). For each candidate Z , check relevance / exclusion / independence and give a verdict:

- 1 Distance to the nearest university at age 18.
- 2 Parental education.
- 3 A lottery that randomly awards tuition waivers among applicants.

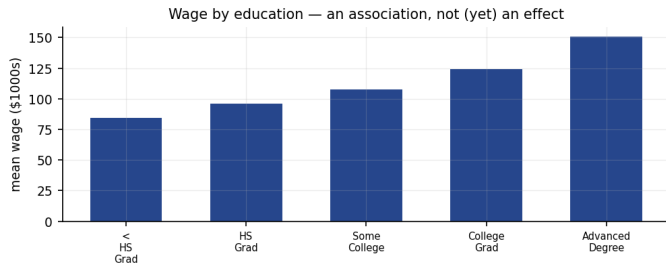
Solution — Exercise A10.5

Solution

Method. Walk the three conditions; one clear failure kills the candidate.

- 1 **Distance:** relevance plausible (nearer $\Rightarrow$ more attendance — check the first stage). Exclusion is the worry: families near universities differ (urban, richer labour markets pay differently). Verdict: *usable only with controls for region — and arguable*. This is the classic Card (1995) design and the classic objection to it.
- 2 **Parental education:** relevant, yes — but it fails independence/exclusion completely: it drives ability, networks and expectations, all of which touch earnings directly. Verdict: *not an instrument; it is a confounder*.
- 3 **The lottery:** relevance testable (waiver $\Rightarrow$ higher enrolment); independence holds by randomisation; exclusion plausible (a waiver moves earnings only via attendance — unless the cash itself matters). Verdict: *the best of the three by far*. Randomisation is what makes instruments credible — which is why the strongest IVs are embedded experiments.

Reality check: Wage data



What we can, and cannot, say

College graduates out-earn non-graduates by \$36.3k; adjusting for age and year barely moves it (\$35.1k). Nothing in this dataset measures ability or family background — so this module's honest verdict is: a large, robust *association*, whose causal share needs a design (a DiD, an instrument, a lottery) the data do not contain.

Extended Exercise A10.1 — Recover the truth in simulation [Python]

Extended exercise (15 min)

Simulate the running example yourself ($n = 2000$, seed 2024): $U \sim N(0, 1)$, $X = U + N(0, 0.5)$, $D = 1\{0.9U + N(0, 1) > 0\}$, $Y = 2 + D + 1.5U + N(0, 1)$.

- 1 Compute the naive difference in means and the regression-adjusted estimate (on X). Compare to 1.0.
- 2 Re-run with $X = U$ exactly (no proxy noise). What happens to the adjusted estimate, and why?
- 3 Vary the proxy noise sd over $\{0, 0.25, 0.5, 1.0\}$ and plot the adjusted estimate against it. What does the curve tell you about “controlling for” badly measured confounders?

Solution — Extended Exercise A10.1 (1/2)

Solution — code

```
import numpy as np
rng = np.random.default_rng(2024)
n, out = 2000, {}
for s in [0, 0.25, 0.5, 1.0]: # proxy noise sd
    U = rng.normal(0, 1, n)
    X = U + rng.normal(0, s, n)
    D = (0.9*U + rng.normal(0, 1, n) > 0).astype(float)
    Y = 2 + D + 1.5*U + rng.normal(0, 1, n)
    naive = Y[D == 1].mean() - Y[D == 0].mean()
    Z = np.column_stack([np.ones(n), D, X])
    adj = np.linalg.lstsq(Z, Y, rcond=None)[0][1]
    out[s] = (naive, adj)
print(f"noise sd {s}: naive {naive:.2f}, adjusted {adj:.2f}")
```

Solution — Extended Exercise A10.1 (2/2)

Solution — reading the output

Part 1. With noise sd 0.5 you reproduce the deck: naive ≈ 2.5 , adjusted ≈ 1.3 – 1.4 (seed-dependent in the second decimal), truth 1.0.

Part 2. With $X = U$ the adjusted estimate sits on ≈ 1.0 : unconfoundedness now holds *exactly*, because the one confounder is fully observed. Adjustment works precisely when its assumption is true.

Part 3. The curve rises smoothly from ≈ 1.0 (no noise) toward the naive value as the proxy degrades — **measurement error in a confounder leaves residual confounding in proportion to the noise**. “We controlled for income” means little if income was self-reported in five broad bins.

Common mistake

Concluding from part 1 that adjustment “does not work”. It did exactly what it promises: removed the bias carried by X . The shortfall is the part of U that X never contained.

Extended Exercise A10.2 — Design an observational study [Integrative]

Extended exercise (15 min)

Your company rolled out a **sales-assistant AI** last year: teams could adopt it voluntarily, and adopters' revenue is now 18% higher. The board asks: "so it caused 18%?" Design the honest analysis:

- 1 Name two plausible confounders and, for each, the *sign* of the bias it induces (use $\gamma\delta$ reasoning).
- 2 You have monthly revenue per team for two years. Which design from this module fits best, and what exactly would you compute?
- 3 The rollout was staggered by region for logistics reasons unrelated to performance. What extra design does that enable?
- 4 List the three robustness checks you would show the board.

Solution — Extended Exercise A10.2

Solution

Part 1. Team quality (good teams adopt tools *and* sell more: $\gamma > 0, \delta > 0 \Rightarrow$ upward bias); desperation adoption (struggling teams grab tools: $\gamma < 0, \delta > 0 \Rightarrow$ downward bias). The 18% could be either side of the truth.

Part 2. Difference-in-differences on the monthly panel: adopters vs never-adopters, change from the pre-adoption year to the post-adoption year — with the pre-period trend plot as the parallel-trends evidence.

Part 3. The staggered, logistics-driven rollout is a candidate **instrument**: region-wave assignment shifts adoption timing (relevance, testable) and is plausibly unrelated to team quality (independence — argue it). Compute the Wald ratio on early- vs late-wave regions.

Part 4. (i) Pre-trend plot; (ii) balance table on observables before/ after matching; (iii) sensitivity: how strong an unmeasured confounder ($\gamma\delta$) would have to be to erase the estimate.

Outline

- 1 The Problem
- 2 Omitted-Variable Bias
- 3 Adjustment & Matching
- 4 Difference-in-Differences
- 5 Instrumental Variables
- 6 Python Lab**
- 7 Summary

Python lab — run it live

Reproduce every number: `make_figures.py`

Every estimate quoted on these slides is computed by `make_figures.py` in this module's folder, from seeded simulations and the bundled `Wage.csv` — run it and read it alongside the deck:

- the running simulation: naive 2.53 / adjusted 1.35 / matched 1.42;
- the from-scratch logistic propensity model and 1:1 matching (ATT);
- the DiD figure and the IV simulation (OLS 2.33 vs Wald 1.08);
- the Wage education gaps (36.3 naive, 35.1 adjusted).

A companion notebook in the course lab format is in preparation; until it lands, the script *is* the lab — it runs top to bottom in the course environment with no extra packages.

The whole module in fifteen lines

```
import numpy as np
rng = np.random.default_rng(2024)
n = 2000
U = rng.normal(0, 1, n) # unmeasured ability
X = U + rng.normal(0, 0.5, n) # what we actually observe
D = (0.9*U + rng.normal(0, 1, n) > 0).astype(float)
Y = 2 + 1.0*D + 1.5*U + rng.normal(0, 1, n)

naive = Y[D == 1].mean() - Y[D == 0].mean() # 2.53
Z = np.column_stack([np.ones(n), D, X])
adjusted = np.linalg.lstsq(Z, Y, rcond=None)[0][1] # 1.35

print(f"truth 1.00 | naive {naive:.2f} | adjusted {adjusted:.2f}")
# adjustment removes the bias carried by X -- and only that
```

Outline

- 1 The Problem
- 2 Omitted-Variable Bias
- 3 Adjustment & Matching
- 4 Difference-in-Differences
- 5 Instrumental Variables
- 6 Python Lab
- 7 Summary

Module A10 in one slide

What we accomplished

Four honest ways to chase a causal effect when nobody randomised — each with its assumption stated out loud.

- Naive comparison = effect + selection: on our simulation, 2.53 for a true 1.0.
- Omitted-variable bias = $\gamma \delta$: sign and size it without data.
- Adjust / stratify / match on observables: $2.53 \rightarrow 1.35\text{--}1.42$ — exactly as far as the measured X reaches, never further.
- Difference-in-differences: time-constant confounding cancels; parallel trends is the bet.
- Instrumental variables: the Wald ratio 1.08 recovers the truth without ever seeing U — if the instrument is truly excludable.

Key formulas at a glance

Omitted-variable bias short = $\tau + \gamma\delta$, with $U = \delta_0 + \delta D + v$.

Unconfoundedness $\{Y(1), Y(0)\} \perp D \mid X$ — the licence for all adjustment methods.

Propensity score $e(x) = \Pr(D = 1 \mid X = x)$; balance on $e(X)$ suffices.

SMD balance check $(\bar{x}_1 - \bar{x}_0) / \sqrt{(s_1^2 + s_0^2)/2}$; aim below 0.1.

DiD $(\bar{Y}_{T,\text{aft}} - \bar{Y}_{T,\text{bef}}) - (\bar{Y}_{C,\text{aft}} - \bar{Y}_{C,\text{bef}})$.

Wald / IV $(\bar{Y}_{Z=1} - \bar{Y}_{Z=0}) / (\bar{D}_{Z=1} - \bar{D}_{Z=0})$ — ITT scaled by the first stage.

Vocabulary checklist

Term	Meaning
confounder	common cause of D and Y ; adjust for it
mediator	lies on the causal path; do not adjust
collider	common <i>effect</i> ; adjusting creates bias
unconfoundedness	all confounders measured — untestable licence to adjust
propensity score	$\Pr(D = 1 \mid X)$; one scalar summarising all covariates
overlap / common support	both groups present at every propensity value
parallel trends	DiD's assumption: equal untreated time-paths
instrument	shifts D ; touches Y only through D ; as-good-as-random
first stage	effect of Z on D ; weak $\Rightarrow$ unstable IV
complier	unit whose D the instrument actually moves

Decision rules of thumb

- Randomised? Use module A1 and stop. Observational? Start from the diagram, not from the regression.
- Adjust for confounders only — never mediators, never colliders, never post-treatment variables of any kind.
- Before matching or weighting: plot propensity **overlap**. After: show the **balance** table. No overlap $\Rightarrow$ no comparison.
- Panel data and a clean untreated group $\Rightarrow$ prefer DiD; always plot pre-trends.
- An instrument beats adjustment *only if* you can argue exclusion out loud; check the first stage before trusting anything.
- Whatever you estimate, attach the sensitivity question: how big a $\gamma\delta$ would undo it?

Common pitfalls

Don't

- 1 Read a regression coefficient on observational data as an effect because the p -value is small.
- 2 Control for everything the dataset offers (mediators and colliders included).
- 3 Match without checking overlap and balance — or report the estimate without them.
- 4 Run DiD without a pre-trend plot.
- 5 Use a clever instrument with a first stage of 0.05.
- 6 Forget that adjustment on a noisy proxy leaves bias in proportion to the noise.

Self-check questions

Quick self-assessment

- 1 The naive gap is 2.53 and the truth 1.0. Which two numbers multiply to the bias, and what are they called?
- 2 Tar deposits in the smoking-disease study: adjust or not, and why?
- 3 Treated: low 5.0 ($n=20$), high 9.0 ($n=80$); control: low 4.0 ($n=80$), high 8.0 ($n=20$). Naive and stratified estimates?
- 4 Control 50 $\rightarrow$ 56, treated 48 $\rightarrow$ 60: the DiD estimate, and the assumption it rests on?
- 5 Wald: $ITT = 0.43$, first stage = 0.40. The IV estimate — and whose effect is it?
- 6 Matching moved the SMD from 1.06 to 0.00 yet the estimate is still 1.42, not 1.0. What explains the gap?

Answers

(1) γ (confounder $\rightarrow$ outcome) $\times \delta$ (group gap in the confounder) — omitted-variable bias. (2) Do not adjust: it is a mediator; you would delete the effect's pathway. (3) Naive $8.2 - 4.8 = 3.4$; stratified 1.0. (4) $12 - 6 = 6$; parallel trends. (5) $0.43/0.40 \approx 1.08$; the compliers'. (6) X is a noisy proxy — balance on X is not balance on U ; residual confounding remains.

Five things to remember

Module A10 in five lines

- 1 **No design, no effect.** A coefficient is causal because of an argument about the world, never because of a fit statistic.
- 2 **Bias** $= \gamma \delta$. Two numbers you can reason about even when you cannot measure them.
- 3 **Adjustment reaches exactly as far as your covariates** — our noisy proxy took 2.53 to 1.4, not to 1.0.
- 4 **DiD trades confounders for one assumption** — parallel trends — and pre-trends are the only evidence for it.
- 5 **An instrument imports randomisation:** Wald $=$ ITT / first stage landed on 1.08 without ever seeing the confounder.

What's next

- **Module A1 — Randomised Controlled Trials** is this module's prerequisite and its benchmark: every method here is an attempt to reconstruct, by argument, what randomisation buys by design.
- **Module A2 — Explainable AI with Shapley Values**: its Pitfall 2 (“an explanation is not a cause”) is precisely this module's subject — now you have the vocabulary to say why.
- **Module A7 — Multiple Testing**: subgroup-hunting for the segment where the effect “works” is a multiplicity problem before it is a causal one.

Appendix — what is in here, and why

Nothing in the appendix is needed to follow the module: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

Contents of the appendix

- **Deriving the OVB formula** — three lines of Chapter 3 algebra; moved out because the result, not the derivation, is what the module uses.
- **ATT vs ATE** — which average the estimators actually target; moved out because the distinction matters only once the main ideas sit.
- **Two-stage least squares** — the regression form of the Wald estimator, for continuous instruments and controls.

Deriving the omitted-variable-bias formula

Long model: $Y = \beta_0 + \tau D + \gamma U + \varepsilon$; auxiliary: $U = \delta_0 + \delta D + v$ with $v \perp D$ by construction. Substitute:

$$Y = \beta_0 + \tau D + \gamma(\delta_0 + \delta D + v) + \varepsilon = \underbrace{(\beta_0 + \gamma\delta_0)}_{\text{new intercept}} + \underbrace{(\tau + \gamma\delta)}_{\text{short-regression slope}} D + \underbrace{(\gamma v + \varepsilon)}_{\text{error, } \perp D}.$$

The composite error is uncorrelated with D , so OLS on D alone estimates $\tau + \gamma\delta$ consistently — consistently *wrong*, by exactly the bias term.

Why this is worth three lines

Every “controlling for X changed my coefficient” moment in Chapter 3 is this identity in motion. It also powers sensitivity analysis: solve $\gamma\delta = \hat{\tau}_{\text{naive}} - 0$ to find how strong a confounder must be to explain your whole estimate.

ATT, ATE, and what each estimator targets

Estimand	Question	Natural estimator
$ATE = E[Y(1) - Y(0)]$	effect for a random unit	stratification weighted by stratum size
$ATT = E[Y(1) - Y(0) \mid D = 1]$	effect for those treated	matching treated to controls
LATE	effect for compliers	Wald / IV

Why the distinction matters

Under treatment-effect *heterogeneity* these differ. Our matching estimator reproduced each treated unit's counterfactual — an ATT. The IV answered for compliers only — a LATE. Neither is wrong; each answers a different policy question (“should we keep the programme for its users?” vs “what if we enrolled everyone?”).

Two-stage least squares (2SLS)

Algorithm

- 1 **First stage:** regress D on Z (and controls X); keep fitted $\hat{D}$.
- 2 **Second stage:** regress Y on $\hat{D}$ (and the same X); the coefficient of $\hat{D}$ is $\hat{\tau}_{2SLS}$.

How to read this

$\hat{D}$ keeps only the part of treatment variation that the instrument explains — the part with no selection in it. With one binary Z , no controls, 2SLS *equals* the Wald ratio exactly. Report the first-stage F : below ~ 10 , the instrument is too weak to trust (our simulation's first stage of 0.40 on $n = 5000$ is comfortably strong).

Common mistake

Running the two stages by hand and using the second stage's printed standard errors — they ignore that $\hat{D}$ was estimated. Real implementations (`linearmodels.IV2SLS`) correct this.